Matrix and FFT exercises

A13-M01 Dimensions and multiplication

Using the slide matrices of sizes 2×3 and 3×2, compute AB and BA, expand c_{21}, and count scalar multiplications for AB.

A13-M02 Block multiplication

Let A be 4×6 and B be 6×5. Split the inner dimension into $2 + 4$. Give block sizes and explain why the two products can be added.

A13-M03 A zero-pivot counterexample

For $[0, 1; 1, 1]$, explain the failure without pivoting, give P, L, U, and check the determinant.

A13-M04 Trace two pivot steps

For the slide 3×3 matrix, list the working matrix after each exchange and elimination. Which identity breaks if the second exchange does not update stored multipliers?

A13-M05 Triangular solves and reuse

Solve for $b = (3, 7, 8)^T$ using the factors. What is the total dense cost for q right-hand sides?

A13-M06 Determinant and one inverse column

Compute the main determinant and the first column of A^{-1} by triangular solves. Compare a complete inverse with one right-hand-side solve.

A13-M07 Verify the complete least-squares fit

For $(0, 1), (1, 2), (2, 2)$, compute $A^T A$, $A^T b$, coefficients, predictions, residuals, and squared error. Verify $A^T r = 0$ and describe residual directions in a scatterplot.

A13-M08 Conditioning and the limits of QR

Prove $\kappa_2(A^T A) = \kappa_2(A)^2$ for full column rank. Quantify the difference for $\mathrm{diag}(1, 10^{-4})$ and explain what QR does and does not guarantee.

A13-X01: A block inverse

For $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ use the top-left entry as B. Compute the Schur complement and block inverse, and verify the identity product. If $B = 0$, can the entire matrix immediately be declared singular?

A13-X02: Positive definiteness and Cholesky

For $A = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix}$, find the Cholesky factor with positive diagonal, prove positive definiteness, and use the factorization to solve $Ax = (6, 7)^T$.

A13-X03: Multiplication from inversion

Given 1×1 matrices $X = [2]$, $Y = [3]$, construct the three-block matrix T and its inverse, then verify by multiplication. In general dimensions, state the cost assumptions for extracting a product from an inverse.

A13-X04: Cost of inversion

Suppose $M(n) = \Theta(n^\omega)$ with $\omega > 2$, and n is a power of two. Solve the SPD block-inversion recurrence and explain why its recursive calls are valid. Explain the reduction for a general real invertible matrix and its limitation in floating-point arithmetic.

A13-X05: Expand Strassen's formula

Using $P_3 = A_{11}(B_{12} - B_{22})$ and $P_5 = (A_{11} + A_{12})B_{22}$, derive $C_{12} = P_3 + P_5$. For $A = [1, 2; 3, 4]$, $B = [5, 6; 7, 8]$, calculate P_3, P_5, C_{12}. Does your derivation still work when block multiplication does not commute?

A13-X06: Rank-deficient least squares

Let $A = [1, 1; 2, 2; 0, 0]$ and $b = (2, 3, 1)^T$. Find all least-squares solutions, their common fitted vector, the residual, and the unique minimum-norm solution. Verify the normal equation.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

A13-X07: Elementwise Cholesky

For $A = [9, 3, 6; 3, 5, 4; 6, 4, 9]$, compute every l_{ij} with the elementwise recurrence and verify LL^T. Explain how the result proves that this particular A is positive definite.

A13-X08: Sufficiency of the normal equation

Let $A = [1, 1; 2, 2; 0, 0]$, $b = (2, 3, 1)^T$, and $\hat{x} = (4/5, 4/5)^T$. Check the normal equation. For arbitrary $z = (u, v)^T$, compute $f(\hat{x} + z) - f(\hat{x})$ exactly and identify equality and all minimizers.

A13-X09: A second pivot and triangular solves

Apply in-place partial-pivot LUP to $A = [2, 0, 1; 1, 0, 2; 0, 3, 1]$. Show both pivot selections, the working matrix after each, the permutation, and L, U. Explain why the second row swap must include the multipliers stored in column 1. For $b = (4, 5, 5)^T$, compute Pb, every forward and back substitution value, and verify the answer.

A13-X10: Singularity and cost

Under exact arithmetic, execute partial-pivot LUP on $A = [1, 2; 2, 4]$ and identify when and why it reports singularity. Does a floating-point pivot below a chosen positive tolerance prove exact real singularity? Under the dense, no-early-exit arithmetic model, give leading flop counts for LUP and Cholesky, and account separately for comparisons and row swaps.

A13-F01

Compute the positive DFT of $[1, 0, 0, 0]$.

A13-F02

For $n = 8$, bit-reverse index 3 and list the bit-reversed order.

A13-F03

Compute both stages for $[1, 2, 0, -1]$, including twiddles.

A13-F04

Prove the inverse formula using the geometric sum.

A13-F05

Recover $[1, 2, 0, -1]$ from $[2, 1 + 3i, 0, 1 - 3i]$ by inverse FFT, showing each stage.

A13-F06

Compute $[1, 2] * [3, 4]$ and explain why $N = 2$ fails.

A13-F07

For lengths 7 and 10, choose the smallest power-of-two FFT size and state the operation bound.

A13-F08

A computed coefficient is 17.49. When can it safely be rounded to 17? Explain what extra evidence is needed.

A13-Z01: Interpolation and field arithmetic

Over $\mathbb{F}_5$, interpolate values $1, 3, 2$ at points $0, 1, 2$ using Lagrange bases. Find the Vandermonde determinant and explain uniqueness.

A13-Z02: Finite-field transform and a counterexample

For $p = 17$ and $N = 4$, show that $\omega = 4$ has order 4, transform $[1, 2, 0, -1]$, and invert the result. Why is $\omega = 16$ unsuitable for length 4?

A13-Z03: Parallel cost and padding

In the stated EREW model with supplied address map and twiddle factors, count the butterflies, work and depth for $N = 8$. Can a length-4 transform directly compute the linear convolution of two length-3 coefficient vectors? Explain.

A13-Z04: Complex convolution by hand

Use forward $\omega = i$ and pad $A(z) = 1 + 2z$, $B(z) = 3 + 4z$ to length 4. List both forward transforms, each pointwise product, all unscaled inverse sums and the final coefficients.

A13-Z05: Follow the invariant and memory

Rearrange a length-8 sequence by three-bit reversal. Which original indices belong to each contiguous block after stages 1 and 2? Explain the transition from stage 2 to stage 3. Compare peak space for sequential depth-first release and retention of every level.

A13-Z06: Construct interpolation and show nonuniqueness

Over $\mathbb{F}_5$, use points $0, 1, 2$ and values $1, 3, 2$. Start with $G(z) = \prod(z - x_i)$, find each $Q_i = G/(z - x_i)$ and $d_i = Q_i(x_i)$, and construct the interpolation coefficients. State the cost for n points. Give two different degree-less-than-3 polynomials that both vanish at 0 and 1.

A13-Z07: Horner invariant

Evaluate $A(z) = 2 + 3z + 4z^2$ at $x = 2$ with Horner's method, listing k and b at each step. Prove the invariant $b = \sum_{j=k}^{2} a_j x^{j-k}$ after processing k. Compare the operation count for one point and for separately evaluating n points.

A13-Z08: Two forward transforms

Take $N = 4$, $\omega = i$ and $x = [1, 2, 0, 0]$. List its DFT, prove that applying the same forward DFT again and dividing by 4 gives $[1, 0, 0, 2]$, and explain how to recover the inverse using either a forward DFT or the inverse root.

References and attribution

CLRS, Introduction to Algorithms, third edition, chapters 28 and 30. Matrix: PDF pages 834–861. FFT: PDF pages 919, 921, 923, 928–936, 939–941, 944–945 (PDF page numbering, third edition). MIT 6.046J Fall 2004 lecture 3 slides, PDF pages 27 and 35–43, and Fall 2005 lecture 3 notes support matrix multiplication and Strassen, not FFT.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study